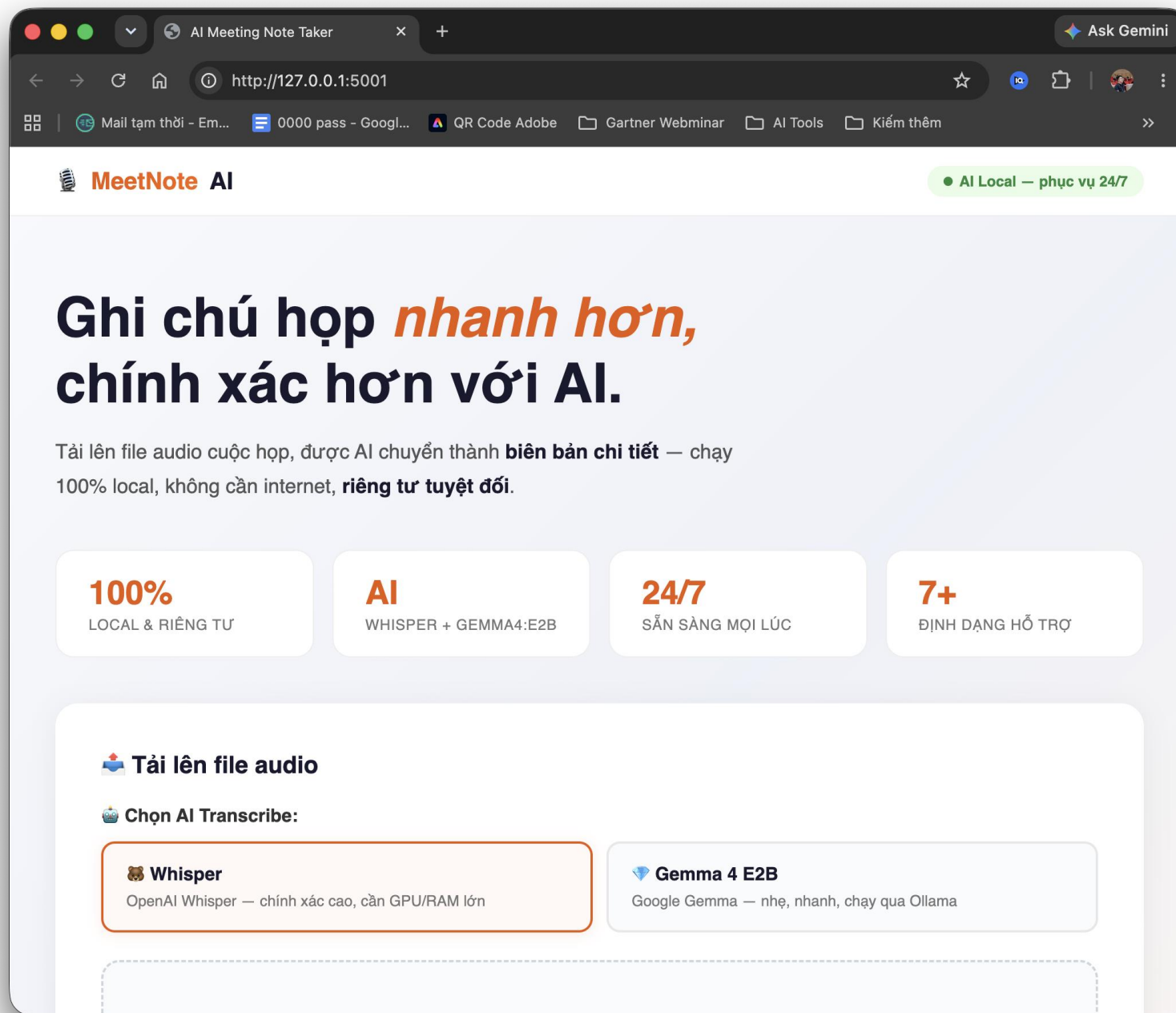


Hướng dẫn code A-Z MeetNote AI

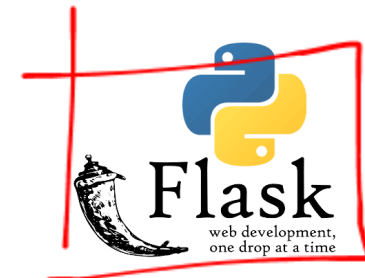
Công cụ chuyển ghi âm
cuộc họp thành meeting
note local 100%



Nguyên vật liệu



Gemma 4



The Gemma 4 Family Matrix: From Mobile to Workstation

Context

Gemma 4 is a family of multimodal open-weights models built by Google DeepMind, designed for advanced reasoning and agentic workflows. The family scales from "Effective" edge-optimized models for mobile devices to "Frontier" models capable of outperforming much larger systems on high-end workstations.

Four Sizes for Every Deployment

Models scale by "intelligence-per-parameter" to fit hardware from smartphones to H100 GPUs.

	E2B (Effective 2B) 	E4B (Effective 4B) 	26B A4B (MoE) 	31B (Dense) 
Architecture	Dense	Dense	Mixture-of-Experts (MoE)	Dense
Context Window	<u>128K Tokens</u>	128K Tokens	256K Tokens	256K Tokens
Key Feature	<u> Native Audio Input</u>	<u> Native Audio Input</u>	 High-Speed Inference	 <u>Frontier Intelligence</u>
Target Hardware	 Mobile / IoT	 Laptop / Mobile	 Consumer GPU	 H100 / Workstation

Universal Capabilities



Native Thinking & Reasoning

All models feature a "thinking mode" for step-by-step logic and complex multi-turn reasoning.



Multimodal & Agentic Out of the Box

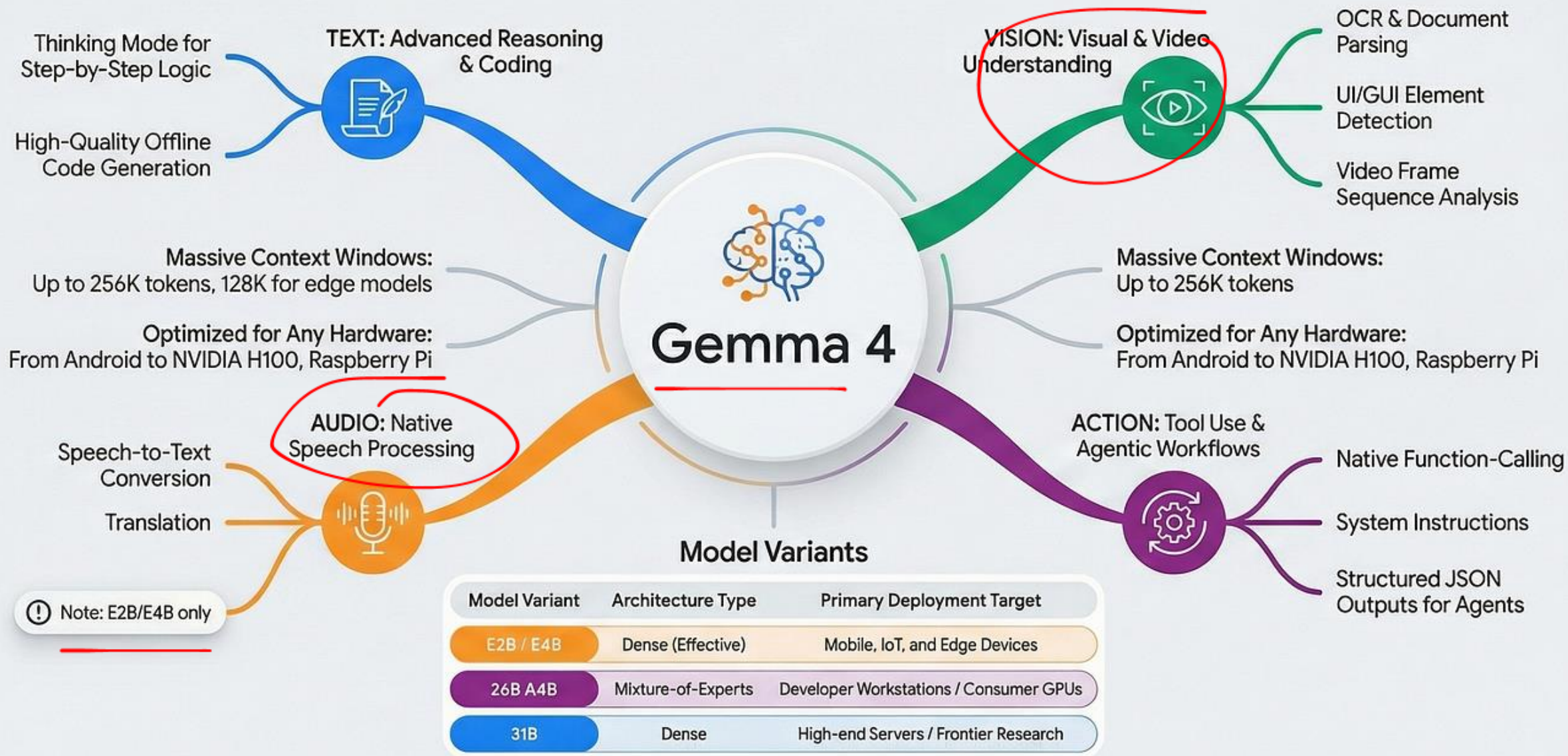
Natively supports image/video processing, function calling, and structured JSON outputs for autonomous agents.

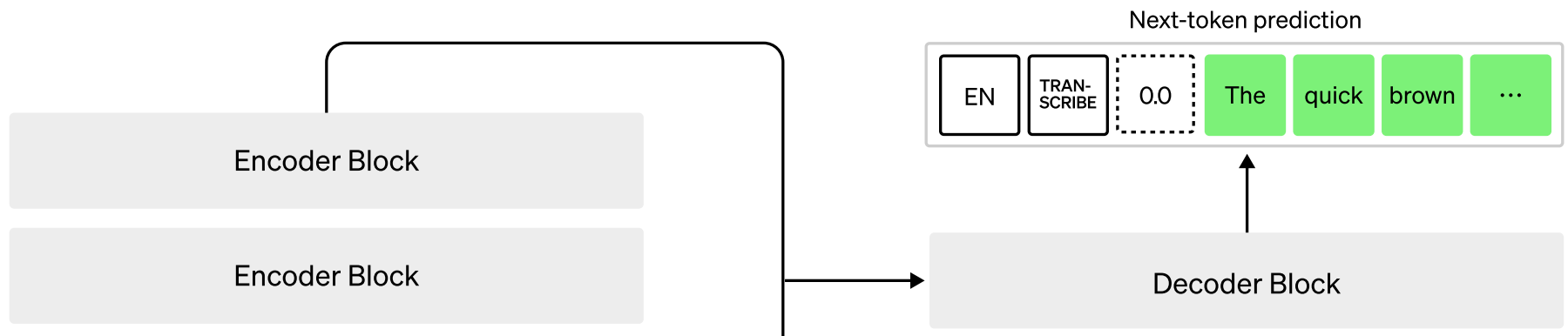


Apache 2.0 Licensing

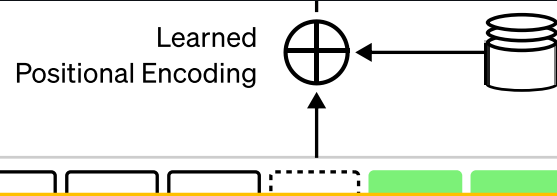
Released under a commercially permissive license for complete developer flexibility and digital sovereignty.

Gemma 4: The Multimodal Capability Map



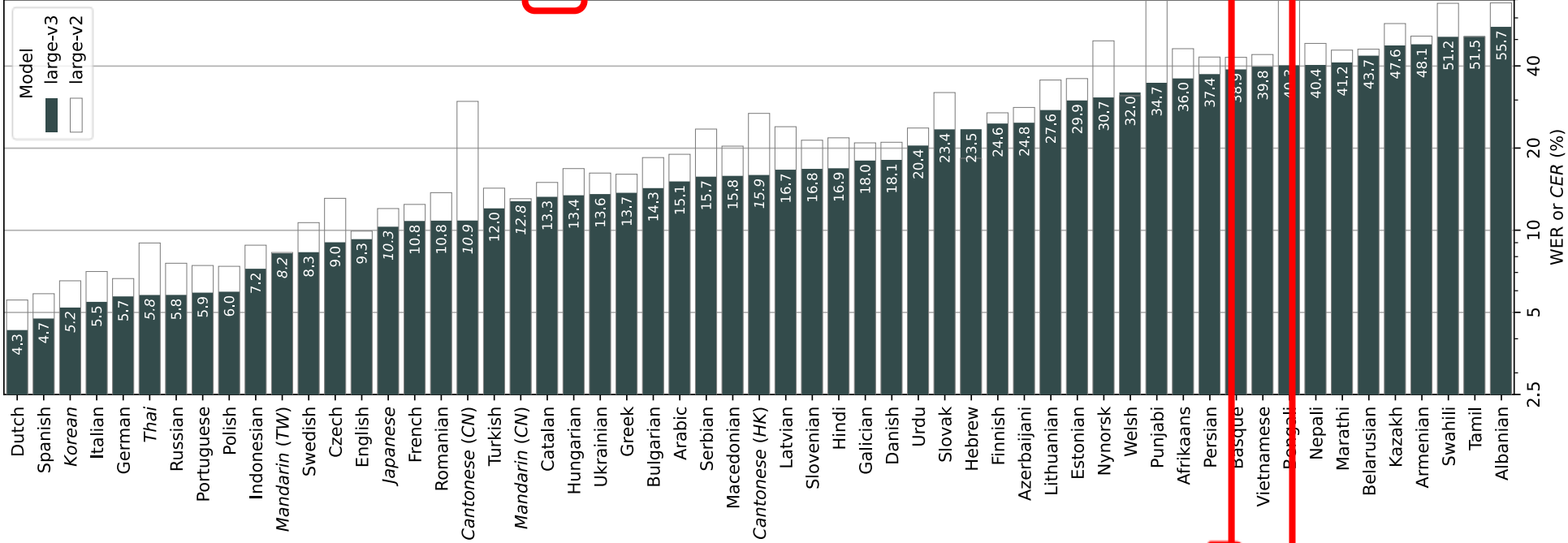


Size	Parameters	English-only model	Multilingual model	Required VRAM	Relative speed
tiny	39 M	<code>tiny.en</code>	<code>tiny</code>	~1 GB	~10x
base	74 M	<code>base.en</code>	<code>base</code>	~1 GB	~7x
small	244 M	<code>small.en</code>	<code>small</code>	~2 GB	~4x
medium	769 M	<code>medium.en</code>	<code>medium</code>	~5 GB	~2x
large	1550 M	N/A	<code>large</code>	~10 GB	1x
turbo	809 M	N/A	<code>turbo</code>	~6 GB	~8x

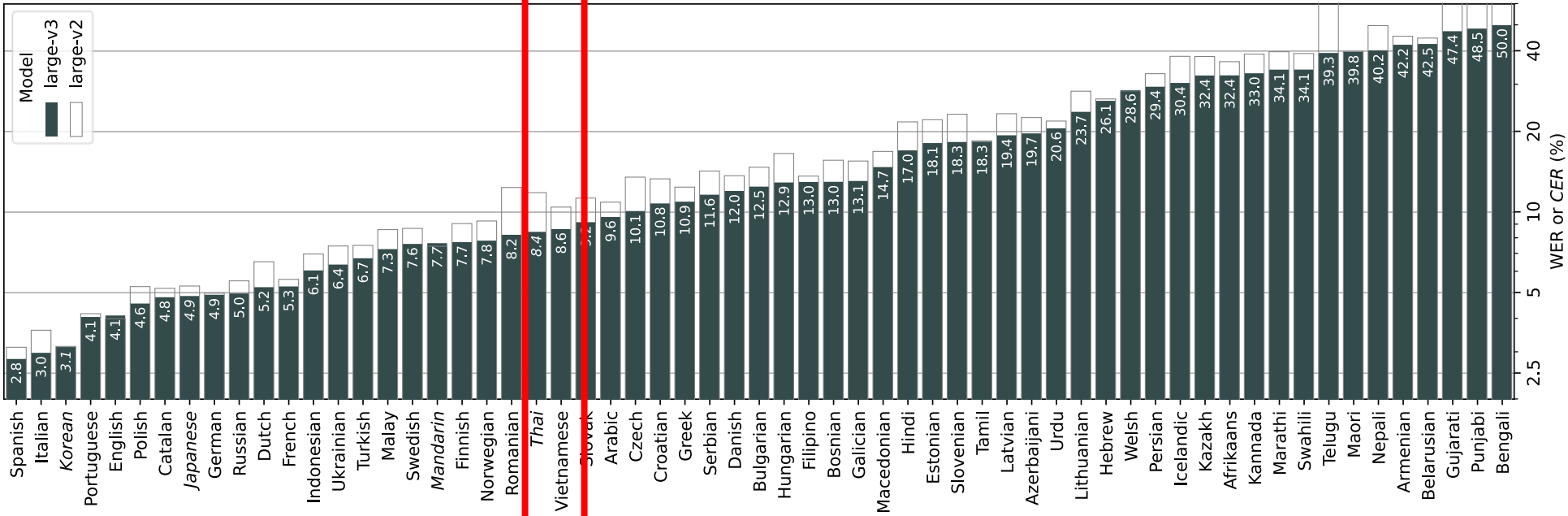


Whisper is a general-purpose speech recognition model. It is trained on a large dataset of diverse audio and is also a multitasking model that can perform multilingual speech recognition, speech translation, and language identification.

Common Voice 15



FLEURS



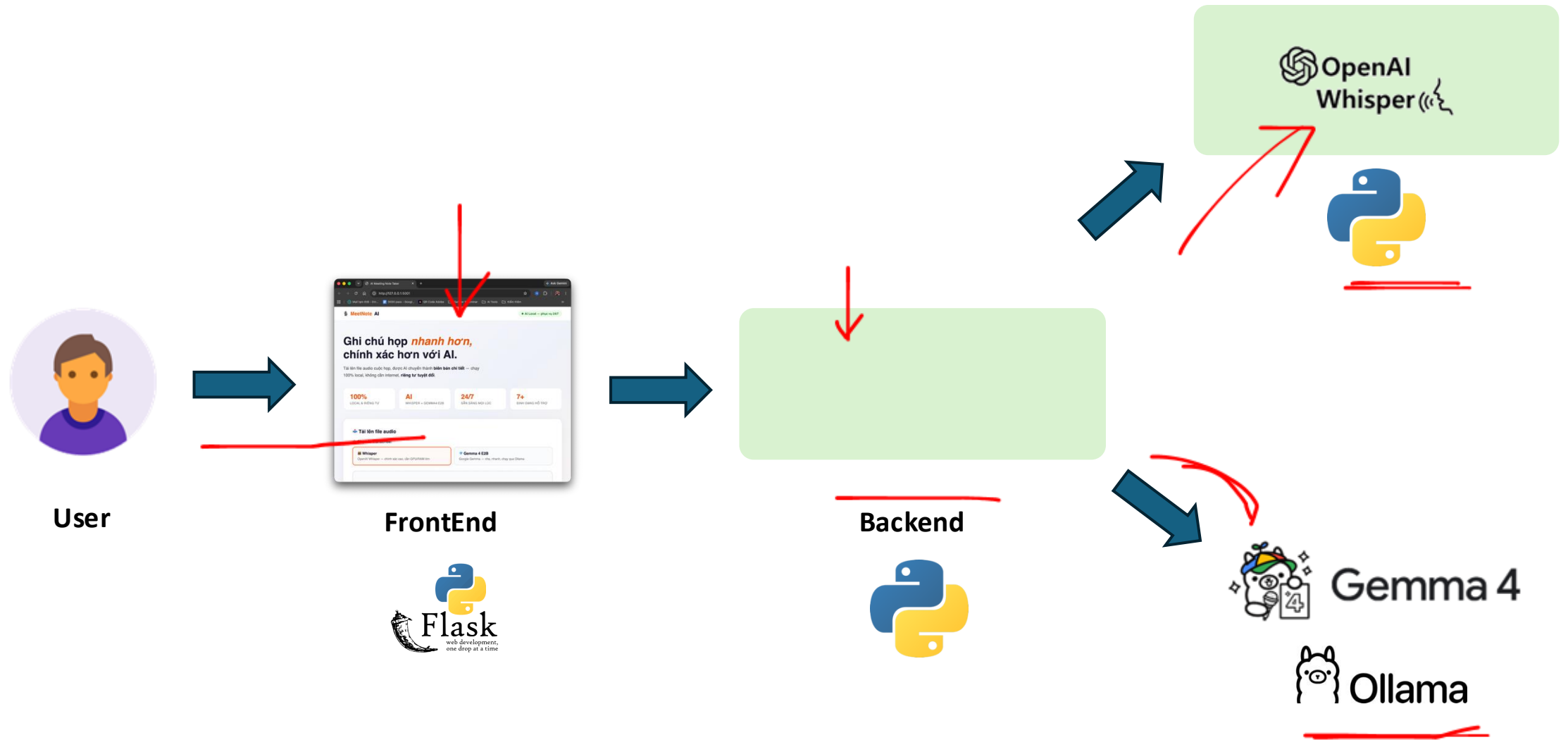


Bảng so sánh tổng quan

Tiêu chí	Ollama	vLLM
Mục tiêu chính	Dễ sử dụng, chạy cục bộ trên máy tính cá nhân.	Hiệu năng cao, phục vụ nhiều người dùng cùng lúc.
Cơ chế xử lý	Xếp hàng yêu cầu (Request Queuing).	Chia lô liên tục (Continuous Batching).
Quản lý bộ nhớ	Cấp phát tĩnh (Static allocation).	Quản lý động qua PagedAttention.
Định dạng mô hình	GGUF (qua Ollama Hub).	HuggingFace nguyên bản (FP16, AWQ, GPTQ).
Hỗ trợ phần cứng	Tốt trên CPU, GPU phổ thông, Apple Silicon.	Tối ưu tuyệt đối cho GPU trung tâm dữ liệu (NVIDIA, AMD).
Thời gian thiết lập	Chỉ vài phút (giao diện CLI hoặc Docker).	Cần cấu hình phức tạp, kiến thức DevOps.



Lắp ráp nguyên liệu



Handons

Run on Macbook Air M4



Cấu trúc thư mục dự án

```
meetnote/
├── app.py                # File chính - Flask web server
├── audio_utils.py       # Cắt audio thành đoạn nhỏ
├── transcribe_whisper.py # Transcribe bằng Whisper
├── transcribe_gemma.py  # Transcribe bằng Gemma 4 E2B
├── summarize.py         # Tóm tắt transcript bằng Ollama
├── templates/
│   └── index.html      # Giao diện web
├── uploads/            # Thư mục chứa file upload tạm
├── temp_segments/      # Thư mục chứa audio đã cắt nhỏ
└── results/           # Thư mục lưu kết quả
```